

Assignment 1

1 Part A: Biased Coin Toss

Theory Questions

1. Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$. Derive the likelihood function and show it is concave in p . (10 marks)
2. Derive the Maximum likelihood estimator for p and compute its variance. (10 marks)

2 Part B: Sampling from a Distribution

Theory Questions

1. Prove correctness of inverse transform sampling. (10 marks)
2. Define importance sampling. Show that for target $\pi(x)$ and proposal $q(x)$: (10 marks)

$$\mathbb{E}_\pi[f(X)] = \mathbb{E}_q \left[f(X) \frac{\pi(X)}{q(X)} \right]$$

3. For $\pi(x) \propto e^{-x^4}$ (10 marks):
 - Explain why direct sampling is difficult
 - Propose a suitable proposal distribution $q(x)$
4. Discuss variance issues in importance sampling. (5 marks)

Coding Questions

1. Perform inverse transform sampling for the Exponential distribution. (10 marks)
2. Implement importance sampling to estimate: (5 marks)

$$\mathbb{E}[X^2], \quad X \sim \pi(x) \propto e^{-x^4}$$

3 Part C: Metropolis–Hastings Algorithm

Theory Questions

1. Derive the Metropolis–Hastings acceptance probability using detailed balance: (10 marks)

$$A(x \rightarrow x') = \min \left(1, \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)} \right)$$

2. Prove that $\pi(x)$ is the stationary distribution. (10 marks)

Coding Questions

1. Implement Metropolis–Hastings for: (10 marks)
 - Standard normal distribution
 - Mixture of Gaussians